FD Implication

Logical implication of FDs – example

name	color	category	dept	price
Gizmo	green	gadget	Toys	$49
Widget	black	gadget	Toys	$59
Gizmo	green	whatsit	Garden	$99

Given:
{name} → {color}
{category} → {dept}
{color, category} → {price}

Logically implies:
{name, category} → {price}

Logical implication rules

Armstrong's Axioms

- Reflexivity: If $B \subseteq A$, then $A \to B$.
- Augmentation: If $A \to B$, then $AC \to BC$.
- Transitivity: If $A \to B$ and $B \to C$, then $A \to C$.

Derivable rules:

- Union: If $A \to B$ and $A \to C$, then $A \to BC$.
- Decomposition: If $A \to BC$, then $A \to B$ and $A \to C$.
- Pseudo-transitivity: If $A \to B$ and $BC \to D$, then $AC \to D$.

Activity: Use FD implications

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Gizmo	green	gadget	Toys	$49
Widget	black	gadget	Toys	$59
Gizmo	green	whatsit	Garden	$99

Reflexivity: If $B \subseteq A$, then $A \rightarrow B$.
Augmentation: If $A \rightarrow B$, then $AC \rightarrow BC$.
Transitivity: If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.
Union: If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$.
Decomposition: If $A \rightarrow BC$, then $A \rightarrow B$ and $A \rightarrow C$.
Pseudo-transitivity: If $A \rightarrow B$ and $BC \rightarrow D$, then $AC \rightarrow D$.

Given:

1. {name} → {color}
2. {category} → {dept}
3. {color, category} → {price}

Logical implication:

1. {name, category} → {price}
2. {name, category} → {name}
3. {name, category} → {color}
4. {name, category} → {category}
5. {name, category} → {color, category}

Rule(s)?

Activity: Derive Union from axioms

Reflexivity: If $B \subseteq A$, then $A \rightarrow B$.
Augmentation: If $A \rightarrow B$, then $AC \rightarrow BC$.
Transitivity: If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.

Union: If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$.